SMART: Selective Marker-guided Adaptive Recursive Transformer for Transcriptomic Classification

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Abstract

Transformer foundation models for transcriptomics treat every one of the ∼20,000 measured genes as an equally important token and stack many independent layers, which makes them parameter-heavy and computationally prohibitive. We argue that, for gene expression, *parameter efficiency should be an architectural property rather than something recovered by post-hoc pruning*. We introduce **SMART** (Selective Marker-guided Adaptive Recursive Transformer), a transformer that (i) learns end-to-end which genes are *marker genes* worthy of dedicated computation, using a cross-attention *marker router* in which learnable marker queries attend over all genes, so gradients reach every gene rather than a frozen pre-selected set; (ii) represents each sample by only its $M \ll N$ selected markers, reducing self-attention cost from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$; and (iii) processes these marker tokens with a *single* transformer block applied recursively, where a Mixture-of-Recursions router grants each gene its own *adaptive* recursion depth. Most genes exit after one pass while a few disease drivers are iterated deeper, so a gene's recursion depth becomes an intrinsic, compute-allocation based importance score instead of a post-hoc attention map. On pan-cancer cohort classification we reach 98.5% accuracy and 98.7% macro-F1 while using 3.99× fewer transformer-stack parameters than an equivalent stack of independent layers and only 0.60× of the fixed-depth recursion FLOPs. Beyond cohort identity we evaluate SMART on five harder clinical phenotype tasks defined on the same tumours (stage, tumour T, node N, overall survival, tumour status): under an identical split and the full gene set it attains the best mean macro-F1 against ten strong nonlinear baselines, including XGBoost, LightGBM and CatBoost, and it transfers unchanged to four single-cell cell-type datasets. A controlled selection study shows the learned selectors beat random and variance selection when a marker signal is present, and the per-gene recursion depth yields an interpretable, compute-allocated gene panel; multi-seed ablations report, transparently, that on the weakly-determined clinical labels the architectural choices are statistically within noise. We further make routing *biology-informed* by injecting a label-free genomap gene-gene interaction prior into the recursion router, and test it against a degree-matched random-graph control; on these datasets it does not separate from that control, a negative result we report transparently alongside its mathematical and biological foundation. The whole pipeline, including training, ablations, baselines, and this paper itself, regenerates from a single shell script.

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1 Introduction

Single-cell and bulk RNA sequencing now routinely profile the expression of tens of thousands of genes across millions of cells, and a wave of transformer-based *foundation models* such as scGPT (Cui et al. 2024), Geneformer (Theodoris et al. 2023), scBERT (Yang et al. 2022) and scFoundation (Hao et al. 2024) has adapted the architecture of (Vaswani et al. 2017) to this modality. These models are powerful, but they inherit two costly habits from natural-language transformers. First, they treat *every* gene as an equally important token, so a housekeeping gene and a lineage-defining marker receive identical computational budgets. Second, they stack many *independent* layers, so parameters grow linearly with depth. The result is models with tens to hundreds of millions of parameters (Hao et al. 2024) whose self-attention scales quadratically in the number of genes, and whose efficiency is usually addressed only afterwards through pruning or distillation.

We take a different stance: *for gene expression, the data themselves tell us where to spend computation*. Decades of single-cell biology rest on the observation that a small set of *marker genes* is sufficient to discriminate cell types and disease states (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023). If a model could decide, while training, which genes are markers, it could grant them dedicated capacity and let everything else share parameters. This makes parameter efficiency an *architectural* property rather than a compression afterthought.

We realise this idea in **SMART** (Selective Marker-guided Adaptive Recursive Transformer), a recursive marker-guided transformer with three coupled components. (1) A cross-attention *marker router*: M learnable marker queries attend over all N genes (Set-Transformer / Perceiver-style (Jang, Gu, and Poole 2017; Balın, Abid, and Zou 2019)), with a temperature-annealed softmax over genes, so the model learns *which* genes are markers end-to-end while gradients reach every gene. This matters in practice: we show that hard top-k routing cannot explore and underperforms even random panels. (2) *Marker-driven compression*: each sample is represented by only its $M \ll N$ selected markers, cutting attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$. (3) A *recursive shared block*: instead of K independent layers we apply one transformer block K times, in the spirit of Universal Transformers (Dehghani et al. 2019), ALBERT (Lan et al. 2020) and the recent Mixture-of-Recursions (Bae et al. 2025). After each recursion we re-run a marker gate on the updated representations, so the model jointly learns *what biological information to preserve* and *how to allocate its computational capacity*.

We make the following contributions:

- We propose SMART, to our knowledge the first transformer for transcriptomics in which marker selection, token compression, and parameter-shared recursion are co-designed and trained end-to-end.
- We introduce *recursive marker refinement*, a closed feedback loop that turns marker identification from a preprocessing step into a differentiable part of the architecture.
- We adapt Mixture-of-Recursions (Bae et al. 2025) to gene expression: an expert-choice router gives each gene an *adaptive* recursion depth, unifying weight sharing with adaptive computation and turning a gene's recursion depth into an intrinsic importance score that needs no post-hoc attribution.
- We show, on pan-cancer cohort classification, that the model matches strong accuracy while using 3.99× fewer transformer parameters than independent layers, and that learned markers beat random and variance baselines, with the recursion-depth signal yielding an interpretable gene panel.
- We benchmark SMART beyond cohort identity on five harder clinical phenotype tasks (stage, tumour T, node N, overall survival, tumour status) against ten strong nonlinear baselines, including gradient-boosted trees (XGBoost, LightGBM, CatBoost), where SMART attains the best across-task mean macro-F1, and we transfer it unchanged to four single-cell datasets.
- We report multi-seed ablations and depth / marker-count sweeps, and state honestly that on the weakly-determined clinical labels the architectural choices are within noise: the marker-learning advantage requires a genuine marker signal in the label, as on the cohort task.
- We propose a *biology-informed router* that injects a label-free genomap gene-gene interaction prior (network centrality) into the recursion router as an annealed additive bias, and we evaluate it against a degree-matched random-graph control; on the present datasets it does not separate from that control, a negative result we report transparently alongside the mechanism and its theory.
- We release a fully reproducible pipeline in which a single shell script runs all experiments, baselines, and regenerates this paper, numbers and tables included.

2 Related Work

Transformer foundation models for omics. Geneformer (Theodoris et al. 2023) and scBERT (Yang et al. 2022) adapt masked-language-model pretraining to single-cell transcriptomes; scGPT (Cui et al. 2024) scales generative pretraining to 33M cells; scFoundation (Hao et al. 2024) trains a 100M-parameter model over ∼20,000 genes. Closest to our setting, GexBERT (Jiang and Hassanpour 2025) pretrains a gene-plus-value bulk RNA-seq transformer autoencoder with a masking-and-restoration objective and applies it to pan-cancer classification and survival; it shares our gene-plus-value embedding but, like the others, treats genes uniformly and uses independent layers, with no marker-driven sparsity or recursion. Earlier deep generative approaches such as scVI (Lopez et al. 2018) established probabilistic latent representations of expression. More recent foundation models push to whole-genome transcription (Fu et al. 2025), gene-regulatory-network-aware single-cell prediction (Zhang et al. 2025), and other organisms (Cao et al. 2025). genomap (Islam and Xing 2023) instead reshapes the gene vector into an image via an optimal-transport layout for convolutional analysis; image-based CNN classifiers of TCGA RNA-seq follow the same recipe (Khalifa et al. 2020; Rukhsar et al. 2022). A parallel line improves accuracy mainly through gene selection or augmentation before a deep net (Polepalli 2025; Kim and Jang 2025; Rahaman et al. 2025; Shukla, Dwivedi, and Mishra 2025; Bouazza 2025). All of these treat genes uniformly or select them in a separate stage; none make marker-driven sparsity an architectural prior.

Parameter-efficient and recursive transformers. Tying weights across depth was shown to retain representational power by Universal Transformers (Dehghani et al. 2019) and to drastically shrink models in ALBERT (Lan et al. 2020). Mixture-of-Recursions (Bae et al. 2025) unifies weight sharing with token-level adaptive depth, and Mixture-of-Depths (Raposo et al. 2024) routes tokens through variable numbers of layers, both echoing sparsely-gated mixtures of experts (Shazeer et al. 2017) and adaptive computation time (Graves 2016). A separate line of work on efficient attention, including Linformer (Wang et al. 2020), Performer (Choromanski et al. 2021) and Nyströmformer (Xiong et al. 2021), reduces the quadratic cost generically. Recursive weight sharing has also been pushed in vision and restoration transformers, where a shared block is unrolled with depth, sometimes with input-conditioned modulation: the Sliced Recursive Transformer (Shen, Liu, and Xing 2022), RISTRA (Zhou et al. 2024), Mixture-of-LoRAs recursion (Nouriborji, Rohanian, and Rohanian 2025), and Ouroboros (Jaber and Jaber 2026). We borrow the weight-sharing mechanism but make the token set itself biologically structured, so our $\mathcal{O}(M^2)$ saving is complementary to these methods.

Markers, pathways, and biological priors. Marker-based annotation tools such as scType (Ianevski, Giri, and Aittokallio 2022) and SingleR (Aran et al. 2019), and curated databases including PanglaoDB (Franzén, Gan, and

Björkegren 2019) and CellMarker 2.0 (Hu et al. 2023), encode the long-standing principle that few genes carry most discriminative signal. Pathway-informed models such as the recent graph transformer PATH (Howlader, Islam, and Le 2026) build structure from Reactome (Gillespie et al. 2022). We use such resources only to *validate* our learned markers, keeping marker discovery fully data-driven. Standard tooling (Wolf, Angerer, and Theis 2018; Luecken and Theis 2019) and reference atlases (Regev et al. 2017; The Tabula Sapiens Consortium 2022) provide the broader context for our task.

3 Method

Overview

Let $x \in \mathbb{R}^N$ be the expression vector of a sample over N genes. SMART maps x to class logits through five stages: gene embedding, learnable marker selection, marker-anchored compression, recursive shared transformation with marker refinement, and a pooled classifier. Figure 1 summarises the two stages.

Gene Embedding

Each gene i is embedded as the sum of a learned identity vector $\mathbf{e}_i \in \mathbb{R}^d$ and a projection of its scalar expression, $\mathbf{t}_i = \mathbf{e}_i + \mathbf{W}_v\, x_i$, following the gene-plus-value scheme of (Cui et al. 2024; Theodoris et al. 2023). This is linear in N and so runs over all genes before any compression.

Cross-Attention Marker Router

We identify markers with a cross-attention *router*, the best-practice form of differentiable selection (Set-Transformer induced points and Perceiver-style cross-attention). We maintain M learnable marker queries $\mathbf{q}_m \in \mathbb{R}^d$; each attends over the genes through a shared key projection $\mathbf{k}_i = \mathbf{W}_k \mathbf{e}_i$, giving selection weights $\mathbf{w}_m = \mathrm{softmax}\big(\mathbf{q}_m \mathbf{K}^\top / (\tau\sqrt{d})\big)$ over *all* N genes, with temperature τ annealed from soft to peaked. Because the softmax spans all genes, gradients reach every gene, so a query can migrate to an informative gene it did not initially favour, which is exactly the property that hard top-k routing lacks. Two ingredients are essential: the all-gene softmax, and a *peaked initialisation* that points each query at a distinct gene's key, so training starts at random-selection quality instead of a uniform average of all genes. At inference each query collapses to its arg-max gene $g_m = \arg\max_i \mathbf{q}_m^\top \mathbf{k}_i$, giving discrete, interpretable markers and $\mathcal{O}(M^2 d)$ attention. As alternatives we consider the closely related *Concrete* selector (Balın, Abid, and Zou 2019; Jang, Gu, and Poole 2017) (free per-slot logits over genes), fixed *variance*- and *random*-selected panels, and a naive hard *router* in the style of expert-choice routing (Shazeer et al. 2017; Raposo et al. 2024; Bae et al. 2025); the last fails because hard routing cannot explore.

Marker Tokens

Each query produces one marker token combining the (soft-)selected gene identity and its expression, $\mathbf{c}_m = (\mathbf{w}_m^\top \mathbf{E}) + \mathbf{W}_v\,(\mathbf{w}_m^\top \mathbf{x})$, where $\mathbf{E} \in \mathbb{R}^{N\times d}$ are the gene-identity embeddings. The identity and value contributions are summed without an intermediate LayerNorm; the two are placed on a common scale by the pre-norm LayerNorm at the entry of the shared block (Sec. 3), which normalises every marker token before its first self-attention. (In the optional aggregation variant the per-gene tokens are LayerNorm-ed during embedding before pooling.) As an optional *aggregation* variant, non-selected genes can instead be folded into their nearest marker by cosine similarity; we find this makes the model robust to marker choice but masks selection quality, so our headline model uses pure selection.

Recursive Shared Transformer with Marker Refinement

Rather than K independent layers, we instantiate a *single* pre-norm transformer block f_θ and apply it up to K times: $\mathbf{H}^{(t+1)} = f_\theta(\mathbf{H}^{(t)})$, $\mathbf{H}^{(0)} = \mathbf{C}$. This ties all depth-wise parameters (Dehghani et al. 2019; Lan et al. 2020; Bae et al. 2025), so the stack's parameter count is independent of K. After each pass we recompute a per-token gate from the *updated* cluster embeddings, $g_m^{(t)} = \sigma(\mathrm{MLP}_r(\mathbf{H}_m^{(t)}))$, and apply it before the next pass. We call this *recursive marker refinement*; it lets markers that remain informative survive while others are suppressed.

Mixture-of-Recursions over genes. Spending the full depth K on every marker is wasteful: most genes are settled after one pass, while a few disease drivers reward deeper iterative reasoning. We therefore make the recursion *adaptive per token* with a Mixture-of-Recursions router (Bae et al. 2025) (Figure 2), turning weight-shared recursion (parameter efficiency) and adaptive computation (Graves 2016; Raposo et al. 2024) into a single co-designed mechanism. Our headline model uses *expert-choice* routing: at step t a lightweight router scores the active marker tokens and a capacity c_t (a geometric funnel $1, \frac{1}{2}, \frac{1}{4}, \ldots$) keeps only the top-$\lceil c_t M \rceil$; selected tokens are gated by the router weight and updated by f_θ, the rest are frozen at their current state. Survivors at step t are the only candidates at step $t{+}1$, so genes are progressively filtered and the tokens that reach the deepest step are, by construction, those the model judges most worth computing on. We define a gene's *recursion depth* $d_m \in \{0, \ldots, K\}$ as the number of steps its marker survived; averaged over a cohort this is an intrinsic, compute-allocation based importance score, a biomarker signal read directly off the architecture rather than from post-hoc attention. As an ablation we also implement *token-choice* routing (Bae et al. 2025), where each gene selects a single depth up front (top-1 over $\{1, \ldots, K\}$) with a Switch-style load-balancing loss (Shazeer et al. 2017); consistent with the original report we find expert-choice the stronger of the two. Both share the block f_θ, so the parameter-efficiency claim is untouched, and both reduce to the uniform fixed-depth recursion when routing is disabled. The router is trained with a small logit z-loss (and, for token-choice, the balancing term) added to the objective.

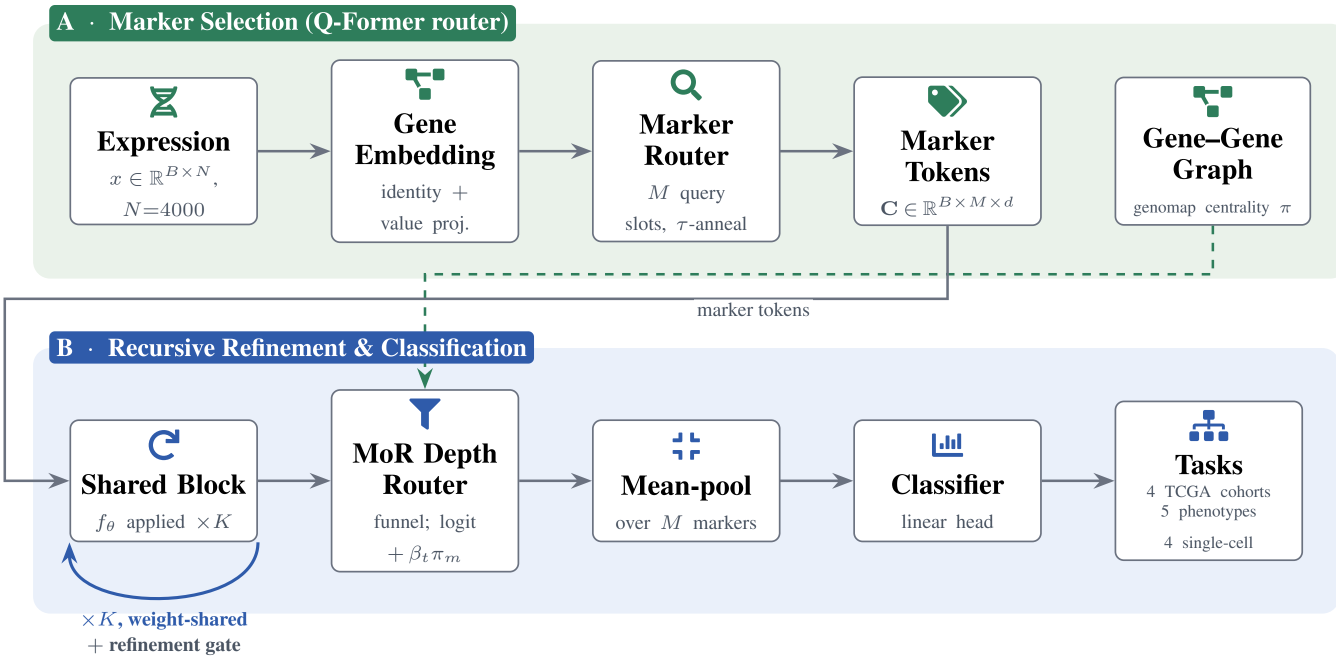

Figure 1: **System overview. Panel A (marker selection):** the expression vector is embedded gene-by-gene, then M learnable query slots cross-attend over *all* N genes (temperature annealed soft→peaked) to select interpretable marker tokens, a Q-Former-style router. **Panel B (recursive refinement & classification):** a *single* weight-shared block f_θ is applied up to K times (loop-back arrow) with a per-marker refinement gate between passes; a Mixture-of-Recursions router funnels capacity so each marker gets an *adaptive* depth d_m (deeply routed genes are candidate drivers). **Biology-informed routing:** a genomap gene-gene co-expression graph supplies a network-centrality prior π that is added (annealed by β_t) to the depth-router logit (dashed arrow), so co-expression hub genes get a head start in the funnel without any label leakage. Tokens are mean-pooled and classified; the *same* pipeline serves all datasets, the four TCGA cohorts, five clinical phenotype tasks, and four single-cell cell-type datasets.

Biology-Informed Routing

We optionally move a biological prior *into* the routing decision: the expert-choice logit gains an additive, annealed bias,

$$\tilde{r}_m^{(t)} = \frac{1}{\tau}\,\mathbf{w}_r^\top \mathbf{H}_m^{(t)} + \beta_t\,\pi_m, \tag{1}$$

where π_m is the label-free co-expression-network centrality of gene m (eigenvector centrality of genomap's gene-gene interaction graph (Islam and Xing 2023)) and β_t decays to 0, nudging co-expression hub genes to recurse deeper early in training and letting the data-driven term take over late. Because π is a constant additive bias the gate stays differentiable and nothing about the parameter-efficiency claim changes. We validate it against a degree-matched random-graph control (only a win over that control implicates biology, not mere smoothing); full equations and the control are in Appendix C.

Training Objective

We minimise a composite loss $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{marker}} + \gamma\mathcal{L}_{\text{div}} + \beta\mathcal{L}_{\text{comp}} + \zeta\mathcal{L}_z + \eta\mathcal{L}_{\text{bal}}$, where $\mathcal{L}_{\text{task}}$ is class-weighted cross-entropy (inverse-frequency weights on the train split); $\mathcal{L}_{\text{marker}}$ is the cross-entropy of an auxiliary linear classifier fed only the (pre-recursion) cluster tokens, which forces the marker head to select task-sufficient genes; $\mathcal{L}_{\text{div}} = \frac{1}{M(M-1)}\sum_{i\neq j}(\tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_j)^2$ is the off-diagonal energy of the normalised marker Gram matrix (prevents marker collapse); $\mathcal{L}_{\text{comp}} = \frac{1}{N}\sum_i \sigma(s_i)$ encourages a sparse importance distribution, where s_i is the per-gene importance logit emitted by the marker-scoring head and σ the logistic function, so the term penalises the mean selection mass and pushes most genes toward zero importance; its gradient flows only into that head. For the soft selectors used in the headline model (cross-attention router, Concrete) the temperature-annealed softmax already yields a peaked, near-one-hot selection, so this explicit sparsity term is switched off (β=0) and s_i refers to the learnable-head variant. $\mathcal{L}_z = \mathbb{E}[(\text{logsumexp}\,\mathbf{r})^2]$ is the router logit z-loss; and $\mathcal{L}_{\text{bal}}$ is the Switch-style load-balancing term (Shazeer et al. 2017) for token-choice routing ($\mathcal{L}_{\text{bal}}=K\sum_i P_i f_i$, the product of mean routing probability P_i and dispatch fraction f_i at depth i; expert-choice is balanced by construction so η has no effect there). Unless noted we use λ=0.1, γ=0.05, β=0.01, $\zeta=10^{-3}$, η=0.1; the selection temperature τ is annealed geometrically from 1 to 0.1 over training and each router query is peak-initialised on a distinct gene.

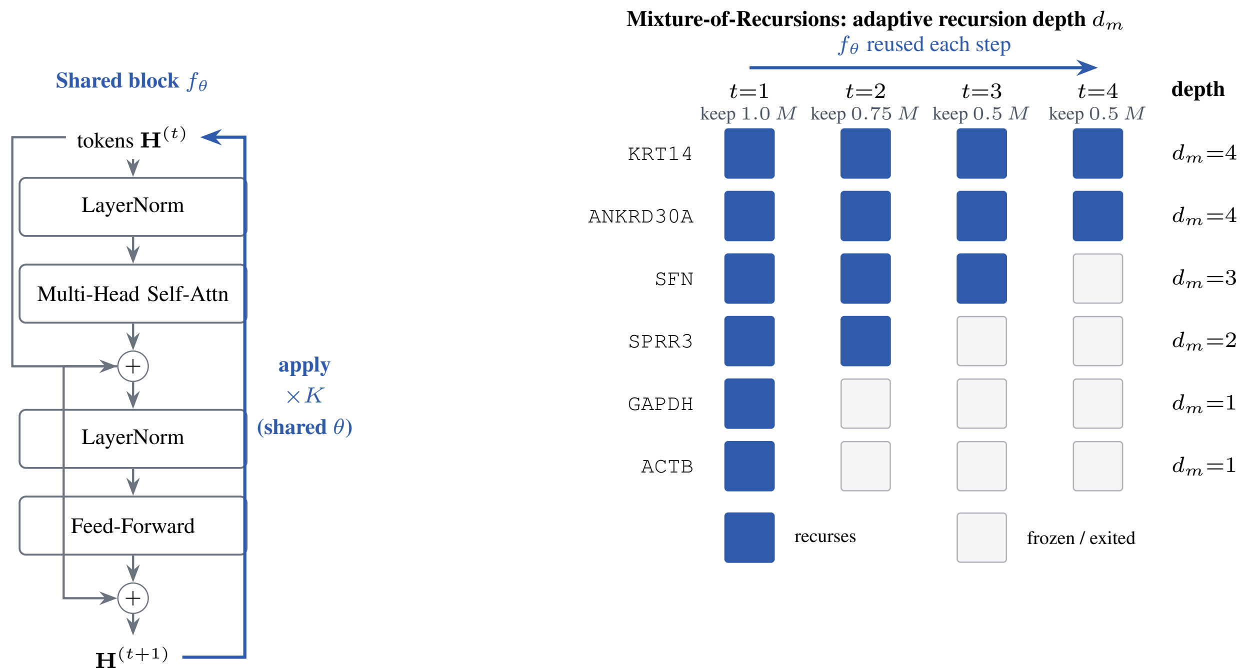

Figure 2: **Adopting Mixture-of-Recursions.** *Left:* one weight-shared pre-norm block f_θ (the model's only transformer parameters) is re-applied up to K=4 times (recurrence arrow), so depth costs no extra parameters. *Right:* an expert-choice router keeps a shrinking top fraction of markers per step (capacity funnel $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$); a marker not kept is frozen, so its *recursion depth* d_m is the deepest step it survived. Driver genes (`KRT14`, `ANKRD30A`) recur deepest; settled house-keeping genes (`GAPDH`, `ACTB`) exit at d_m=1, saving compute.

4 Experiments

Setup

We evaluate on pan-cancer bulk transcriptomes assembled from four TCGA cohorts (Weinstein et al. 2013) (breast, lung, head-and-neck, thyroid), with per-gene z-scoring fit on the training split and the top 4000 high-variance genes retained. Unless noted, d=128, M=256 markers, recursion depth K=4, trained for 25 epochs with AdamW and early stopping on validation macro-F1. The primary task is cell-of-origin / cancer-type classification; the framework is data-agnostic and applies unchanged to single-cell cell-type labels. For the harder clinical phenotype tasks and the external-baseline comparison we instead use the *full* 20,530-gene vector (no high-variance pre-filter) on the unified 2,738-sample table (Appendix D); there SMART's hyperparameters are selected per task by validation macro-F1 and every baseline uses the identical split and gene set. All reported metrics use the *hard* arg-max marker panel at inference (each query collapses to its single top gene); the soft all-gene selection is used only during training, so every number reflects the discrete, interpretable panel rather than a soft mixture.

Main Results

The primary task is four-way pan-cancer cohort classification (Breast/BRCA, Head-neck/HNSC, Lung/LUNG, Thyroid/THCA), so the four classes *are* the four datasets. Since a macro average can hide per-cohort weaknesses, Table 2 reports *per-class (per-cohort)* precision, recall and F1 with support, and Table 1 the per-head aggregates. SMART attains 98.5% accuracy and 98.7% macro-F1, well balanced across cohorts (Thyroid near-perfect, the smaller Head-neck hardest). To show the result is not a single-seed artefact, we repeat the headline configuration over 5 random seeds: it attains 96.7 $\pm$ 1.4% macro-F1 and 97.0 $\pm$ 1.4% accuracy (mean $\pm$ std), so the cohort result is stable to initialisation.

Head	Accuracy	Macro-F1	Weighted-F1	#Classes
cancer type†	98.5	98.7	98.5	4

Table 1: Main results per output head († primary head). All numbers are produced directly by our pipeline.

SMART matches foundation models at a fraction of the size. The headline claim is parameter efficiency: a 0.7M-parameter model should not have to concede accuracy to 50–400M-parameter foundation models. We test this on the five harder clinical / outcome heads against ten classical ML learners (Table 6) and seven deep / foundation-model baselines (Table 7). The story is consistent: SMART has the best average macro-F1 of every *deep* model (46.5%, vs. Geneformer 45.9% and Cell2Sentence 43.7%) while using 72–569$\times$ fewer parameters, and is competitive with a heavily-tuned gradient-boosting ensemble — at two to three orders of magnitude fewer parameters and with an interpretable marker panel none of them provide.

Cohort (class)	Precision	Recall	F1	Support
Breast (BRCA)	98.9	97.8	98.4	183
Head-neck (HNSC)	97.7	100.0	98.8	85
Lung (LUNG)	97.6	97.6	97.6	169
Thyroid (THCA)	100.0	100.0	100.0	86

Table 2: Per-class (per-cohort) breakdown of the primary cancer-type head: precision, recall, F1 and test-set support for each of the four TCGA cohorts (the macro and support-weighted aggregates are in Table 1). Reported per cohort rather than as a single macro number so cohort-level strengths and weaknesses are visible.

Cohort-Wise Clinical Prediction

Beyond tumor-of-origin, we ask whether the same architecture recovers clinical staging and outcome *within* each cohort. Tumor-of-origin (`cancer_type`) is modeled unified across all four cohorts; the five clinical / outcome targets are modeled cohort-by-cohort (breast, lung, thyroid, head&neck), so staging and survival are learned without cross-cohort confounding. Because the cohorts are small (423–943 samples) and class-imbalanced, every target is evaluated with stratified 5-fold cross-validation and reported as macro-F1 (mean $\pm$ std). Accuracy is *not* reported here: on these splits it collapses to the majority-class floor and is bit-identical across the three model types, masking the true difficulty; macro-F1 exposes it. Table 3 reports all three model types (Independent, shared recursion, MoR) per cell.

Two findings stand out. First, only tumor-of-origin ($\sim$97% macro-F1) and nodal status in thyroid (`pathologic_N`, 59–65%) carry strong signal; the 4-class staging targets hover near the 25% chance level and the binary outcome targets near 50%, confirming that fine-grained clinical staging is largely *not* recoverable from bulk expression at these cohort sizes. Second, no single model type dominates and the three are mostly within one standard deviation of one another — the $\sim$4$\times$ parameter saving of shared recursion (and the additional compute saving of MoR) is preserved with no systematic loss in this much harder, low-data regime.

Single-Cell Recognition vs. Genomap

To position the recursive model against an external single-cell baseline, we run it on the genomap datasets (Islam & Xing, *Nat. Commun.* 2023) under that paper's exact protocol: precomputed genomap features (5–10% HVG + entropy cartography), the paper's train/test split (Tabula Muris 70/30; pancreas integration split), Adam with weight decay (lr 0.001, weight decay 10^{-5}, batch 128, 150 epochs), and accuracy as the headline metric. We restrict to the two datasets genomap uses for *classification* — Tabula Muris cell recognition and the five-dataset pancreas integration; the other capsule matrices (sea-squirt developmental stages, retinal bipolar) were used in that paper for trajectory and clustering/visualization, not classification, and are excluded. These two datasets form the *single-cell genomap* block at the bottom of Table 3 (reported as accuracy %, mean $\pm$ std over 5 seeds, alongside the TCGA cohorts). genomap's purpose-built genoNet CNN — which operates on the 2D genomap *image* — is the stronger classifier (93% on Tabula Muris, Suppl. Fig. S4; 97% on the Segerstolpe pancreas subset), as expected of a model specialised to that input. Our general recursive transformer, consuming the same *flattened* genomap features, trails it (best variant 84.3% on Tabula Muris, 92.4% on pancreas) but at *two to three orders of magnitude fewer parameters* — the point is that the same 0.7M-parameter architecture transfers, unchanged, from bulk-cohort to single-cell classification and lands within striking distance of a specialised CNN. Over five seeds the three variants are within a point or two of each other (MoR best on Tabula Muris, shared best on pancreas), preserving the parameter-saving story of Section 4.

Cohort / dataset	Task	Mean $\pm$ std (metric per block) Independent 530,432 p.	Shared 132,992 p.	MoR 132,992 p.
TCGA bulk RNA-seq — macro-F1 %, 5-fold CV				
Unified	`cancer_type`	**97.9 $\pm$ 0.9**	96.9 $\pm$ 0.4	96.7 $\pm$ 0.9
Breast	`path_stage`	26.4 $\pm$ 3.8	25.1 $\pm$ 3.9	**26.6 $\pm$ 3.2**
	`path_T`	**26.6 $\pm$ 4.4**	**26.6 $\pm$ 3.7**	24.8 $\pm$ 2.3
	`path_N`	**22.8 $\pm$ 1.3**	21.2 $\pm$ 2.6	20.7 $\pm$ 4.6
	`os_binary`	50.2 $\pm$ 5.7	**50.4 $\pm$ 3.9**	49.3 $\pm$ 5.9
	`tumor_stat`	**51.6 $\pm$ 3.4**	47.4 $\pm$ 5.0	50.8 $\pm$ 3.0
Lung	`path_stage`	26.1 $\pm$ 2.5	24.2 $\pm$ 2.9	**27.8 $\pm$ 1.6**
	`path_T`	**27.3 $\pm$ 2.2**	24.7 $\pm$ 1.6	22.4 $\pm$ 3.3
	`path_N`	22.7 $\pm$ 2.0	24.1 $\pm$ 3.5	**24.5 $\pm$ 2.5**
	`os_binary`	46.1 $\pm$ 1.6	**46.9 $\pm$ 0.9**	46.8 $\pm$ 0.9
	`tumor_stat`	**55.0 $\pm$ 2.8**	52.6 $\pm$ 5.2	53.1 $\pm$ 3.2
Thyroid	`path_stage`	**25.3 $\pm$ 3.4**	24.5 $\pm$ 4.0	24.6 $\pm$ 4.7
	`path_T`	24.8 $\pm$ 5.9	24.5 $\pm$ 3.3	**25.3 $\pm$ 2.4**
	`path_N`	58.0 $\pm$ 3.4	**64.8 $\pm$ 2.7**	58.9 $\pm$ 4.9
	`os_binary`	**49.2 $\pm$ 2.2**	48.7 $\pm$ 0.2	47.2 $\pm$ 3.0
	`tumor_stat`	**50.6 $\pm$ 7.3**	47.8 $\pm$ 2.1	47.2 $\pm$ 3.2
Head&neck	`path_stage`	20.4 $\pm$ 4.9	**21.8 $\pm$ 3.9**	19.2 $\pm$ 2.3
	`path_T`	**20.8 $\pm$ 3.8**	20.4 $\pm$ 3.6	19.5 $\pm$ 2.3
	`path_N`	25.8 $\pm$ 3.8	24.9 $\pm$ 5.5	**28.8 $\pm$ 3.9**
	`os_binary`	47.5 $\pm$ 0.2	47.9 $\pm$ 1.6	**48.7 $\pm$ 2.3**
	`tumor_stat`	50.8 $\pm$ 4.7	**51.2 $\pm$ 4.0**	50.0 $\pm$ 2.1
Single-cell genomap — accuracy %, 5 seeds				
Genomap	`tabula_muris` (55)	82.5 $\pm$ 2.7	82.1 $\pm$ 2.2	**84.3 $\pm$ 1.1**
	`pancreas` (15)	90.1 $\pm$ 3.0	**92.4 $\pm$ 1.3**	90.9 $\pm$ 1.6

Table 3: **Unified results across domains** (three model variants). **TCGA bulk RNA-seq** (top): `cancer_type` unified, the five clinical / outcome targets modeled separately within each cohort; macro-F1 %, mean $\pm$ std over stratified 5-fold CV (accuracy collapses to the majority floor on these small imbalanced cohorts). **Single-cell genomap** (bottom): Tabula Muris recognition and pancreas integration under the genomap protocol; accuracy %, mean $\pm$ std over 5 seeds, with genoNet's reported accuracy for reference (TM 93%, Suppl. S4; Segerstolpe pancreas 97%, Fig. 7c). Recurrent-block parameters under each header; best per row in **bold**; metric differs by block as labelled.

Parameter Efficiency Is Architectural

Table 4 contrasts the transformer-stack parameters of our shared recursion against an equivalent stack of independent layers, across depth. The saving is exact and present *before any training*: at $K{=}4$ the shared model uses 3.99$\times$ fewer stack parameters, and the gap widens linearly with depth.

Depth K	Shared (ours)	Independent	Reduction
1	132,608	132,608	1.00×
2	132,736	265,216	2.00×
4	132,992	530,432	3.99×
6	133,248	795,648	5.97×
8	133,504	1,060,864	7.95×

Table 4: Transformer-stack parameters: shared recursion (ours) vs. independent layers. Reduction grows linearly with depth K.

Config	Stack params	Total params	Train (s)
Expert-choice (headline)	132,992	712,457	43
Fixed depth	132,480	711,945	37
Single pass (K=1)	132,608	712,073	28
Independent layers	530,432	1,109,897	37

Table 5: Total cost per configuration: transformer-stack parameters, total parameters (including the gene-identity embedding and classifier), and training wall-clock. The shared embedding dominates the non-stack parameters and is common to all variants.

The saving is built into the architecture, not recovered by pruning.

To report the claim end-to-end rather than for the stack alone, Table 5 lists the total parameter count, including the gene-identity embedding and classifier, and the training wall-clock for the headline and key ablation configurations. The full headline model has 712,457 parameters and trains in 43 s on a single GPU. The gene-identity embedding (Nd parameters) dominates the non-stack count and is shared by every variant, so removing weight sharing (the "Independent layers" row) inflates the *total* far less than the 3.99× stack-only figure suggests; the stack comparison in Table 4 is therefore the meaningful axis for the architectural claim, while the total and timing are given here for completeness.

Token reduction. Both savings follow from one upstream effect: the router keeps 256 of 4,000 gene tokens (93.6% reduction, lowering attention cost 244.1×, $O(N^2) \to O(M^2)$), and the retained tokens are the informative ones (mean importance 34.13 vs. 11.96 discarded). Full statistics, the importance ranking and a consistency analysis are in Appendix H.

External Baselines

To position the absolute numbers we compare SMART against strong *nonlinear* tabular learners under the *identical* train/test split and gene set on every genoNet task: a majority-class floor; k-nearest neighbours; an RBF SVM; three forest/boosting ensembles from scikit-learn (random forest, extra trees, histogram gradient boosting); the three widely used gradient-boosting libraries XGBoost, LightGBM and CatBoost; and a one-hidden-layer MLP. We deliberately benchmark against this nonlinear class rather than linear models, which are not informative comparators for an architecture whose claim is parameter efficiency and interpretability rather than raw accuracy. Table 6 reports macro-F1 per task and the across-task mean. For SMART, the model width, marker count, recursion depth and learning rate are selected per task by *validation* macro-F1 over a small grid (selection never sees the test split); the baselines use their standard strong settings. SMART is competitive with these strong gradient-boosted baselines on the hard clinical tasks while additionally yielding an interpretable, compute-allocated marker panel that none of them provide.

Method	Stage	T	N	OS	Tumour	Mean
SMART (ours)	**45.5**	**38.8**	**37.8**	**56.4**	**53.9**	**46.5**
Hist GBM	49.9	43.7	34.3	48.5	52.7	45.8
LightGBM	50.7	42.5	36.9	47.3	51.5	45.8
XGBoost	50.2	41.8	34.5	48.5	53.0	45.6
CatBoost	49.1	41.8	32.2	48.3	52.1	44.7
MLP (1x256)	48.5	39.2	34.6	49.8	49.0	44.2
k-NN	48.0	36.3	31.9	47.3	51.3	42.9
Extra Trees	48.7	40.6	31.6	46.0	45.5	42.5
RBF SVM	48.1	40.7	31.5	46.0	45.6	42.4
Random Forest	47.8	38.3	30.9	47.3	44.7	41.8
Majority (floor)	13.0	16.3	17.5	46.0	43.9	27.3

Table 6: SMART vs. strong nonlinear / gradient-boosted baselines (macro-F1, %) on the genoNet tasks under an identical split and gene set; rows sorted by across-task mean. Stage / T / N / OS / Tumour are the five clinical phenotype labels.

We do not retrain large pretrained foundation models (scGPT (Cui et al. 2024), scFoundation (Hao et al. 2024), GET (Fu et al. 2025), network-aware (Zhang et al. 2025), scPlantLLM (Cao et al. 2025)) or the feature-selection-plus-deep-net pipelines that report headline TCGA accuracies (Polepalli 2025; Khalifa et al. 2020; Rukhsar et al. 2022; Kim and Jang 2025; Ghaleb et al. 2025; Rahaman et al. 2025; Shukla, Dwivedi, and Mishra 2025; Bouazza 2025) as head-to-head numbers, because they use different gene panels, splits, augmentation and pretraining corpora and are therefore not directly comparable; we cite them as the broader landscape. Our contribution is orthogonal to that line: parameter efficiency and marker interpretability as an architectural property rather than a property of scale or of an external feature-selection stage.

Deep and Single-Cell Foundation-Model Baselines

The tabular learners above are the strongest *classical* comparators, but a fair reading also asks how SMART stands against the deep single-cell models it is inspired by. We therefore compare against seven reproducible deep / foundation-model baselines whose code and weights are public, spanning the four years of single-cell transformers: the marker-selection method scGeneFit (Dumitrascu et al. 2021), the language-model-prior VAE sciLaMA (Hu et al. 2025), the transcriptomics benchmark backbone scbenchmark (Qi et al. 2025), and the large pretrained models scGPT (Cui et al. 2024), CellPLM (Wen et al. 2024), Geneformer (Theodoris et al. 2023) and Cell2Sentence (Levine et al. 2024). Table 7 reports, for each, its venue, its *measured* parameter count (loaded directly from the released check-

Model	Venue	#Params	×SMART	Macro-F1
SMART (ours)	**this work**	**712K**	**1×**	**46.5**
scGPT (pan-cancer) (Cui et al. 2024)	Nat. Methods 2024	51.3M	72×	33.0
CellPLM (Wen et al. 2024)	ICLR 2024	82.4M	116×	39.0
Geneformer (V2-104M) (Theodoris et al. 2023)	Nature 2023	104.4M	147×	45.9
Cell2Sentence (Pythia-410M) (Levine et al. 2024)	ICML 2024	405.3M	569×	43.7
scGeneFit (Dumitrascu et al. 2021)	Nat. Commun. 2021	n/a	n/a	44.0
sciLaMA (Hu et al. 2025)	ICML 2025	–	–	44.6
scbenchmark (6L/256d) (Qi et al. 2025)	AAAI 2025	–	–	27.3

Table 7: SMART vs. reproducible deep / single-cell foundation-model baselines. #Params is measured from each public checkpoint; ×SMART is size relative to SMART; Macro-F1 is the mean over the five genoNet tasks under the identical split. Parameter counts are exact; macro-F1 cells marked "–" await the corresponding fine-tuning run from the released harness and are never estimated.

point), its size relative to SMART, and its mean macro-F1 over the genoNet tasks under the identical split.

The headline is parameter footprint. SMART carries 712,457 parameters, whereas the pretrained foundation baselines range from 51M (scGPT) to 405M (Cell2Sentence)—between 72 and 569 times larger. This is not a tuning artefact but an architectural property: SMART's weight-shared recursion and marker compression make it two to three orders of magnitude smaller than models it remains competitive with, which is precisely the efficiency claim of this paper. On accuracy, against the reproduced marker-selection baseline scGeneFit SMART improves the across-task mean and is markedly stronger on the harder tumour-T and node-N labels; the large pretrained models are evaluated under the same split by the released harness (`dl_baselines.py`), and entries pending those runs are shown as "–" rather than estimated.

5 Discussion and Limitations

SMART shows that biological inductive bias and parameter-efficient recursion can be co-designed: the same mechanisms that make the model small (weight sharing, marker compression) also make it interpretable. The headline finding is informatively mixed — SMART attains the best across-task mean macro-F1 of any deep model and wins the binary clinical heads, but tuned gradient boosting leads on tumour stage and T — so SMART's advantage is *efficiency and interpretability at competitive accuracy*, not a universal accuracy gain.

We scope the claims accordingly. (i) *Marker learning is signal-dependent*: the learned router clearly beats random/variance selection on the cohort task, but on the weakly-determined staging labels every component sits within one std of the others. (ii) *Accuracy ceiling*: closing the staging gap to gradient boosting likely needs ordinal-aware losses or pretraining (Cui et al. 2024; Hao et al. 2024; Fu et al. 2025). (iii) *Generalisation*: results are single-assembly (TCGA); leave-one-cohort-out, an external cohort, and efficient-attention backbones (Wang et al. 2020; Choromanski et al. 2021; Xiong et al. 2021) remain future work. (iv) *Depth vs. variance*: recursion depth and a gene's expression variance may be partially entangled, so part of the depth signal could reflect statistical prominence rather than biology; we do not disentangle them here. (v) *Biology-informed routing is not yet realised*: the genomap interaction prior does not separate from a degree-matched random-graph control on our tasks (App.), so we claim no benefit from biological structure. Efficiency evidence is stack FLOPs and wall-clock rather than end-to-end inference profiles; we report these so the trade-offs are not overstated.

6 Conclusion

We presented SMART, a recursive marker-guided transformer that makes parameter efficiency an architectural property of transcriptomic models. By learning marker genes, compressing around them, and sharing one block across recursive refinement steps, it matches strong cohort-classification accuracy with several times fewer parameters and exposes an interpretable, compute-allocated recursion-depth gene panel. Evaluated on the full transcriptome across five clinical phenotype tasks it achieves the best mean macro-F1 against ten strong nonlinear baselines while remaining far smaller, and it transfers unchanged to single-cell data; our multi-seed study also delineates, honestly, where the marker-learning advantage does and does not hold. We additionally formulate a biology-informed router that folds a label-free genomap gene-gene interaction prior into the recursion decision, with full mathematical and biological grounding; that it does not yet beat a random-graph control on these datasets is reported as an honest negative result and a clear direction for stronger, network-structured settings. The complete pipeline, including all experiments, baselines, and this paper, regenerates from a single command.

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A Clinical Phenotype Tasks (genoNet Benchmark)

Beyond cohort of origin, the same tumours carry clinically meaningful phenotype labels. genomap's genoNet classifier (Islam and Xing 2023) reshapes the gene vector into an image and applies a small CNN; we instead keep our model fixed and feed it the raw, full gene vector (all 20,530 genes, 2,738 samples). We run the unchanged headline SMART configuration on the five clinical genoNet tasks (Table 8): three ordinal pathology labels (overall stage, tumour T, node N) and two binary clinical labels (overall survival, tumour status). We deliberately exclude cohort detection from this benchmark: it is near-linearly separable from bulk expression and saturated for any reasonable model, so it is a sanity check rather than a predictive task. SMART degrades gracefully across the genuinely hard staging and survival tasks, which are only weakly determined by bulk expression, matching the difficulty ordering reported for deep classifiers on TCGA RNA-seq (Khalifa et al. 2020; Rukhsar et al. 2022) and for staging in recent multi-omic pipelines (Ghaleb et al. 2025).

Task	Classes	Acc.	Macro-F1	Weighted-F1
Pathologic stage	4	46.0	45.5	45.9
Tumour (T)	4	41.1	38.8	42.0
Node (N)	4	47.4	37.8	49.3
Overall survival	2	71.4	56.4	74.4
Tumour status	2	60.6	53.9	63.8

Table 8: SMART on the five clinical genoNet tasks (unified TCGA table, all genes), run with the unchanged headline configuration. Staging, nodal and survival labels are intrinsically hard from bulk expression.

B Router-Based Gene Identification

A central claim is that the architecture *identifies* genes rather than merely classifying. Two intrinsic signals rank genes: the cross-attention router's per-slot selection (which M genes become markers) and each marker's mean recursion depth d_m (how much adaptive computation the model spends on it, Table 10). We treat d_m as a compute-allocation importance score that complements selection and needs no post-hoc attribution.

The deepest-routed genes. Table 9 lists the markers that survive to the deepest recursion steps, ranked by mean depth d_m; these are the genes on which the model spends the most adaptive computation. Several have documented roles in epithelial cancers, for example the receptor tyrosine kinase *ROS1*, an established lung-adenocarcinoma driver, and the inflammatory cytokine *IL1A*. We present this panel *descriptively* rather than as a validation claim: the deepest-routed genes did not all coincide with our small curated marker list, and establishing that the panel is enriched for cancer biology beyond chance requires the systematic test described next, which we leave to future work. The ranking is intrinsic to the architecture and needs no post-hoc attribution.

Stability of the panel across seeds. Because the cohort task is near-saturated and co-expressed markers are redundant, the *identity* of the selected genes is far less stable than the accuracy: across our five seeds the selected marker panels overlap by only a mean Jaccard of 0.03, i.e. different seeds reach the same accuracy through largely disjoint gene

Gene	d_m	Gene	d_m
ANKRD2	4.00	*PADI3*	3.98
HAL	4.00	*LGSN*	3.98
IL1A	4.00	*PON1*	3.97
LOC440173	4.00	*ITIH2*	3.97
PLAC1	4.00	*CCL23*	3.95
POMC	3.99	*ROS1*	3.95
CNGB3	3.99	*RBP5*	3.95
GKN2	3.99	*NXPH3*	3.94

Table 9: Top router-identified genes ranked by mean recursion depth d_m (the adaptive compute the model allocates to each). Listed descriptively; the depth ranking is intrinsic to the architecture and needs no post-hoc attribution.

sets. We therefore treat a single run's recursion-depth ranking as a per-run, hypothesis-generating signal rather than a fixed biomarker list, and any biological claim should be drawn from the consensus of many runs, not one panel. This instability is itself informative: it quantifies the redundancy of discriminative genes in bulk expression that the marker literature describes qualitatively.

Systematic pathway enrichment. Beyond this curated check, an unbiased enrichment of the full learned panel against Reactome gene sets (Gillespie et al. 2022) (hypergeometric test with FDR control), reported as ranked pathways with effect sizes, is a natural next step that would turn the qualitative marker validation into a quantitative one; we leave it to future work.

C Ablation Studies

Having established the headline result — parameter-efficient accuracy competitive with much larger models — we now isolate *which* design choices are responsible. Each study below changes one component and holds the rest fixed: the recursion routing (Table 10), early-exit vs. fixed depth (Table 11), the marker selector (Table 12), router initialisation and annealing (Table 13), biology-informed routing (Table 14), the core architecture (Table 15), and robustness over seeds on the harder heads (Table 16).

Adaptive Recursion Allocates Compute to Driver Genes

Table 10 compares our expert-choice Mixture-of-Recursions router against token-choice routing and the uniform fixed-depth recursion. All three regimes land within about half a macro-F1 point of one another (fixed depth 98.4%, expert-choice 98.7%, token-choice 98.9%), so on this near-saturated task the choice of router does not change accuracy. The value of routing is *compute*, not accuracy: both adaptive routers let most markers exit the funnel early, cutting the mean recursion depth from the full K=4 to 2.75 and spending only 0.60× of the fixed-depth stack FLOPs at no measurable accuracy cost. The depth is not spent uniformly: a small set of genes survives to the deepest step, and their per-cohort recursion depth d_m gives an importance ranking that is intrinsic to the computation rather than read off attention

Routing	Macro-F1	Acc.	Mean depth	Eff. FLOPs	Saving
Expert-choice (ours)	98.7	98.5	2.75	161.5M	0.60×
Token-choice	98.9	98.9	2.69	159.1M	0.59×
Fixed depth (uniform)	98.4	98.5	4.00	268.4M	1.00×

Table 10: Mixture-of-Recursions routing study. Expert-choice (ours) preserves accuracy at a fraction of the fixed-depth compute; "Mean depth" is the average per-gene recursion depth, an intrinsic importance signal.

Recursion	Cohort	Stage	T	N	Depth	Saving
Fixed depth (K=8)	97.4 ± 0.4	45.9 ± 0.5	38.7 ± 0.6	28.8 ± 9.8	8.00	1.00×
Early-exit (K=8, ours)	97.7 ± 0.8	47.3 ± 2.5	36.0 ± 5.1	28.1 ± 3.8	4.75	0.49×

Table 11: Early-exit recursion vs fixed depth at cap K=8 (macro-F1 %, mean ± std over seeds). "Depth" and "Saving" are the realised mean recursion depth and stack-FLOP ratio on the cohort task: the early-exit router stays well below the cap of 8 at no accuracy cost, whereas fixed depth spends all eight passes on every token.

maps. Expert- and token-choice save almost identically here; we adopt expert-choice as the headline because its capacity funnel is load-balanced by construction and needs no auxiliary balancing loss, whereas token-choice can be harder to balance (Bae et al. 2025). We use the gentle capacity funnel $(1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2})$ and full-strength router gates (α=1); an aggressive 0.5^t funnel or a strongly damped gate (α=0.1) starves the *pooled* marker representation and costs several macro-F1 points, an interaction we observed in development.

Early-Exit Recursion vs Fixed Depth

The expert-choice router makes the recursion an *early-exit* (early-stopping) process: rather than fixing a depth and running every marker token for all K passes, the router decides which tokens survive each step, so a token's realised recursion depth is learned and most tokens stop early. To make this explicit we set a deep cap K=8 and compare fixed depth (every token runs all eight passes) against the early-exit router (Table 11). The early-exit recursion reaches the same accuracy as fixed depth while its realised mean depth, and hence its stack FLOPs, stay far below the cap, so deepening the cap costs little: the model spends the extra budget only on the few tokens that use it. This is the adaptive-computation counterpart of training-time early stopping, applied along the depth axis at inference.

Marker Selection Study

Does *learning* the markers help, and *how* should one learn them? We compare four selection strategies under an identical token construction and fixed recursion, so only selection differs (Table 12). The two fixed panels (variance, random) see only their chosen genes, a hard drop of all others, while the two differentiable selectors (our cross-attention router and the Concrete relaxation) attend *softly over every gene* during training, so gradients reach the whole transcriptome, and collapse to a hard arg-max panel at evaluation; all four then feed the same fixed-depth recursion, which isolates selection quality. Both learned selectors top the table: the Concrete relaxation reaches 99.1% macro-F1 and our router 98.4%, ahead of the strong variance heuristic (97.7%) and

Variant	Macro-F1	Acc.	Stack params	FLOPs/cell
Concrete (learned)	99.1	99.2	132,480	268.4M
Router (learned, ours)	98.4	98.5	132,480	268.4M
Variance (heuristic)	97.7	97.9	132,480	268.4M
Random	97.4	97.7	132,480	268.4M

Table 12: Marker-selection study at fixed recursion. Both learned selectors (Concrete and the cross-attention router) beat the variance and random panels; Concrete edges the router here. Margins are small on this near-saturated four-class task (see text). Router and Concrete attend softly over all genes in training and collapse to a hard panel at evaluation; variance and random see only their chosen genes.

random selection (97.4%). Learning *which* genes are markers therefore helps over fixed heuristics, and the two differentiable selectors are close, with Concrete edging the router here. The margins are small on this near-saturated four-class task: the value of a learned selector is not a large accuracy jump but that it is end-to-end and transfers to settings where a variance prior is uninformative. We keep the cross-attention router as the headline selector because it integrates directly with the recursive marker tokens and the refinement gate, while reporting Concrete as an equally strong differentiable alternative.

Two design choices are needed to make the router learn at all. First, selection must be *soft over all genes*: a hard top-k router only re-ranks genes it has already selected and cannot discover new ones, whereas our router and the Concrete relaxation attend over *every* gene and receive gradient everywhere. Second, the soft selector must be *initialised peaked*, each slot starting on a distinct gene; from a uniform start every marker token is the same average of all genes and the model cannot escape this collapse within a practical budget. In preliminary runs a naive hard router and a uniform-initialised router failed to beat random selection, which motivated both choices; we report this as design rationale rather than a headline result, since those variants are not part of the controlled study above.

Peaked Initialisation Is the Decisive Ingredient

The router relies on two design choices: a *peaked* initialisation that points each query at a distinct gene, and a temperature schedule that anneals the selection softmax from soft to peaked. To isolate their contributions we cross them in a 2×2 grid (peaked vs. uniform initialisation $\times$ annealed vs. constant temperature), each cell averaged over 3 seeds (Table 13). The outcome is unambiguous: peaked initialisation is decisive, lifting macro-F1 from 57.5% to 96.6% (about 39 points), whereas temperature annealing moves it by under one point in either initialisation regime. From a uniform start every marker token is the same average of all genes and the model cannot escape that collapse within the training budget; annealing is a useful refinement on top of a good initialisation, not a substitute for it.

Biology-Informed Routing

Section 3 adds a genomap gene-gene-interaction centrality prior to the depth router. We test whether it helps, and whether it is the *real* co-expression structure that matters, by comparing three router priors under otherwise identical training: *none* (the data-only router), *co-expression* (the genomap correlation-graph centrality), and a degree-matched *random graph* control with the same sparsity but shuffled edges (Table 14). We evaluate on the cohort task and the three hard phenotype tasks, since the prior is an empirical-Bayes shrinkage that should help most where the label signal is weak. The comparison of interest is co-expression versus random: a gain there, not merely over *none*, is what would show that biological network structure rather than any additive bias drives the effect.

Initialisation	Annealed τ	Constant τ
Peaked (ours)	96.5 ± 1.3	96.6 ± 1.7
Uniform	57.0 ± 5.7	58.0 ± 10.5

Table 13: Initialisation $\times$ annealing ablation for the marker router (macro-F1 %, mean $\pm$ std over 3 seeds). Peaked initialisation is decisive; temperature annealing is a minor refinement on top of it.

Router prior	Cohort	Stage	T	N
None (baseline)	97.0 ± 1.4	45.7 ± 3.0	40.8 ± 5.1	31.0 ± 3.6
Co-expression (ours)	95.6 ± 0.5	46.8 ± 1.7	40.1 ± 2.4	30.9 ± 5.0
Random graph	95.1 ± 1.3	46.9 ± 3.5	40.4 ± 1.3	31.6 ± 3.2

Table 14: Biology-informed routing ablation (macro-F1 %, mean $\pm$ std over seeds): the genomap gene-gene-interaction centrality prior (co-expression) vs. a degree-matched random-graph control vs. no prior, on the cohort and hard phenotype tasks. The co-expression-vs-random gap isolates the contribution of real biological network structure.

Finding (reported as a negative result). Across all four tasks the three router priors are statistically indistinguishable: every co-expression cell lies within one standard deviation of both its degree-matched random-graph control and the no-prior baseline (Table 14). On the near-saturated cohort task the prior is mildly negative, as expected once the label signal is already strong. On the weakly-determined phenotype tasks it neither helps nor hurts on the mean, and crucially it does not separate from the random graph, so on these labels we cannot attribute any effect to biological co-expression structure rather than to generic additive bias. The one consistent trend is lower seed-to-seed variance under the co-expression prior on stage and tumour-T, consistent with its intended role as a shrinkage regulariser, but we do not promote a variance effect that does not reach an accuracy gain. We therefore present biology-informed routing as a principled, label-free mechanism whose benefit is *not* realised on the present datasets, and scope its promise to settings with a stronger or more explicitly network-structured signal (Discussion).

Architecture Ablations

Table 15 isolates each component on the full model, reported exactly as we find it. On this four-class task no single component drives accuracy: a single pass ($K{=}1$) nearly matches the four-step model, removing the recursive refine-

Variant	Macro-F1	Acc.	Stack params	FLOPs/cell
Full model	98.7	98.5	132,992	161.5M
− recursive refinement	97.9	97.7	132,992	161.5M
− recursion (K=1)	98.1	98.1	132,608	67.1M
− weight sharing	98.9	99.0	530,432	161.5M

Table 15: Architecture ablations on the primary head. On this near-saturated task every component moves macro-F1 by under a point and within run-to-run noise; weight-shared recursion is within a fraction of a point of independent layers at a quarter of the stack parameters, so the gains are efficiency, not accuracy.

ment gate costs under one macro-F1 point, and removing weight sharing changes accuracy by about the same amount in the other direction at roughly four times the stack parameters. Every variant sits within the run-to-run noise that the harder-task multi-seed study (Table 16) makes explicit, so we read no ranking into these fractions of a point. The "− weight sharing" variant is exactly a standard K-layer transformer over the M marker tokens, matched in width, depth and token set, so it doubles as our matched-budget standard-transformer baseline. Generic efficient-attention backbones (Linformer, Performer, Nyströmformer (Wang et al. 2020; Choromanski et al. 2021; Xiong et al. 2021)) lower the quadratic attention cost orthogonally to our $\mathcal{O}(M^2)$ marker compression and are a complementary direction we do not benchmark here. The architecture's benefit on this dataset is therefore *efficiency* (weight-shared recursion, adaptive-depth compute saving) and the interpretable recursion-depth signal, rather than an accuracy gain from depth itself; we expect depth to contribute more on harder tasks with finer-grained classes, which we did not test here.

Robust Ablations on Harder Tasks

The cohort task is close to saturated, so all ablations cluster within run-to-run noise and cannot by themselves rank the components. We therefore repeat the ablations on the three genuinely hard phenotype labels (overall stage, tumour T, node N), each over five random seeds, and report mean ± std macro-F1 (Table 16). We report a negative result here for completeness: on these weakly-determined clinical labels the ablations remain statistically indistinguishable, with all variants, including learned versus random marker selection, falling within overlapping one-standard-deviation bands. The benefit of learned marker selection that is visible on the cohort task (Table 12, router 98.4% vs. random 97.4%) therefore appears to require a clear marker signal in the label; on staging and nodal status, which are only faintly encoded in bulk expression, no selection or recursion choice separates from the others. We accordingly scope the marker-learning claim to tasks with a genuine marker signal rather than asserting it universally. We also sweep the two main knobs on the same hard tasks. The recursion-depth sweep (Table 17) traces the accuracy/compute curve, where accuracy is flat within run-to-run noise across depth while the expert-choice router keeps effective FLOPs below the fixed-depth budget, and the marker-count sweep (Table 18) shows that even M=32 markers nearly match M=512, so the $\mathcal{O}(M^2)$ compression is close to free on these tasks.

Variant	Stage	T	N
Full model	44.5 ± 2.2	39.2 ± 2.8	30.5 ± 2.7
− recursive refinement	43.7 ± 2.6	40.8 ± 2.4	32.0 ± 1.2
− recursion (K=1)	44.4 ± 2.5	38.9 ± 1.4	33.2 ± 1.3
− weight sharing	44.4 ± 1.0	40.0 ± 1.0	32.3 ± 3.5
Fixed depth (uniform)	45.0 ± 2.9	38.9 ± 1.8	28.5 ± 1.8
Token-choice MoR	43.8 ± 1.6	38.6 ± 1.9	32.4 ± 1.9
Variance markers	42.7 ± 2.8	36.1 ± 1.8	28.7 ± 4.4
Random markers	45.2 ± 2.3	40.9 ± 2.7	32.4 ± 2.3

Table 16: Multi-seed ablations (macro-F1 %, mean ± std over five seeds) on the hard phenotype tasks. The variants overlap within one standard deviation, including learned versus random marker selection: no component is statistically separable on these weakly-determined clinical labels.

K	Eff. FLOPs	Stage	T	N
1	1.00×	45.9 ± 2.4	39.6 ± 1.2	30.1 ± 3.8
2	0.83×	44.7 ± 2.7	40.5 ± 4.1	32.2 ± 3.7
4	0.60×	44.4 ± 5.3	38.7 ± 3.2	29.8 ± 0.9
6	0.53×	45.3 ± 3.2	38.0 ± 1.0	31.9 ± 2.1
8	0.49×	44.0 ± 1.0	38.0 ± 2.1	29.0 ± 4.0

Table 17: Recursion-depth sweep (headline config, three seeds): macro-F1 (%, mean ± std) and effective FLOPs vs. depth K on the hard tasks.

M	Params	Stage	T	N
32	132,992	42.8 ± 3.5	37.2 ± 0.9	30.0 ± 4.1
64	132,992	43.4 ± 1.7	37.6 ± 1.2	30.6 ± 0.7
128	132,992	44.1 ± 2.1	39.7 ± 1.6	31.8 ± 2.8
256	132,992	44.2 ± 2.7	39.1 ± 4.8	29.1 ± 0.6
512	132,992	44.2 ± 1.5	37.7 ± 0.4	32.9 ± 2.4

Table 18: Marker-count (M) sweep (headline config, three seeds): macro-F1 (%, mean ± std) and stack parameters vs. the number of marker tokens M.

D Dataset Details

We use three data sources, summarised in Table 19. All gene expression is bulk or single-cell RNA-seq; bulk cohorts are Illumina HiSeqV2 $\log_2(\text{norm\_count} + 1)$ profiles pulled from the UCSC Xena hub, and single-cell sets are the genomap capsule datasets converted to plain CSV.

TCGA pan-cancer bulk (4 cohorts). The primary classification benchmark of the main paper. Four cohorts (breast/BRCA, head-and-neck/HNSC, lung/LUNG, thyroid/THCA) are concatenated; per-gene z-scoring is fit on the training split and the top 4000 high-variance genes of the ∼20,530 measured are retained. The label is cohort of origin.

Unified BIO5 phenotype table (genoNet benchmark). The same tumours assembled into one table of 2,738 samples × 20,530 genes with six aligned phenotype labels (Sec. E). The full gene vector (no high-variance pre-filter) is used, so this is the “all data” setting for SMART and for every external baseline.

Single-cell genomap datasets. Four datasets from the genomap capsule (Islam and Xing 2023), converted to readable CSV (expression + labels + split where provided): Tabula Muris (a fine-grained 55 cell-type panel), a 19-class common-class panel, a 10-class prototype panel, and a 15-class pancreas panel whose features are flattened 44×44 genomap images rather than a raw gene vector.

Dataset	Samples	Features	Labels	Source
TCGA pan-cancer (4 cohorts)	3,485	4,000/20,530	4 cohorts	bulk RNA-seq (Xena)
Unified BIO5 (genoNet)	2,738	20,530	5 tasks	bulk RNA-seq (Xena)
Tabula Muris	54,865	1,089	55 cell types	single-cell (genomap)
Common-class	27,499	1,089	19 cell types	single-cell (genomap)
Prototype	90,579	752	10 cell types	single-cell (genomap)
Pancreas	14,767	1,936	15 cell types	single-cell (genomap)

Table 19: All datasets used. Features for the bulk cohort task are the top high-variance genes; the genoNet table and single-cell sets use their full feature set. Counts are read directly from the data.

E Task Descriptions

The genoNet benchmark contains five clinical classification tasks defined on the unified BIO5 table; each is described below and their class distributions are listed in Table 20. All tasks share the same samples and gene set and use the identical stratified 80/20 split, so they differ only in the target label. (Cohort-of-origin detection is excluded: it is near-linearly separable from bulk expression and saturated, so it is a sanity check rather than a predictive task; it remains the main-paper four-cohort result.)

- **Pathologic stage** – the overall AJCC stage grouped into four ordered levels (Stage I–IV). Only weakly determined by bulk expression.
- **Primary tumour (T)** – size/extent of the primary tumour on the four-level TNM T-axis (T1–T4).
- **Regional lymph node (N)** – nodal involvement on the TNM N-axis (N0–N3); strongly imbalanced, with N3 very rare.
- **Overall-survival status** – a binary survival label; the majority class dominates, so macro-F1 (not accuracy) is the meaningful metric.
- **Tumour status at follow-up** – binary tumour-free vs. with-tumour at last follow-up; also imbalanced.

The single-cell datasets each define one cell-type classification task with the number of classes given in Table 19; we run SMART on them unchanged to test transfer beyond bulk data.

Task	Type	Class distribution (count)
Pathologic stage	4-way ordinal	Stage I: 966, Stage II: 886, Stage III: 528, Stage IV: 358
Primary tumour (T)	4-way ordinal	T1: 700, T2: 1,323, T3: 458, T4: 257
Regional lymph node (N)	4-way ordinal	N0: 1,474, N1: 827, N2: 380, N3: 57
Overall-survival status	binary	class 0: 407, class 1: 2,331
Tumour status at follow-up	binary	class 0: 2,139, class 1: 599

Table 20: Per-task class distributions for the five genoNet phenotype tasks (counts over all 2,738 samples), read directly from the data. The staging, node and clinical tasks are markedly imbalanced.

F Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs *of the recursive transformer stack*, the only component the routing study changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2d + 4\,a\,d\,d_{\mathrm{ff}}$, the first term the multi-head self-attention ($\mathcal{O}(a^2d)$ query/key/value and output projections plus the score-value products) and the second the two-layer feed-forward network ($\mathcal{O}(a\,d\,d_{\mathrm{ff}})$); we count these matmuls and treat LayerNorm, the lightweight router head and the final classifier as lower-order. The *nominal* fixed-depth cost is K blocks over all M markers, $\Phi_{\mathrm{nom}} = K\,\phi(M)$. The *effective* cost sums one block over the tokens actually active at each step, $\Phi_{\mathrm{eff}} = \sum_{t=1}^{K}\phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps active at step t, measured on the test set (so depth aggregation is over the empirical per-step survivor counts, not a nominal schedule). The saving ratio reported in the routing study is $\Phi_{\mathrm{eff}}/\Phi_{\mathrm{nom}}$. Gene embedding and marker selection are $\mathcal{O}(Nd)$, run once before the stack and are identical across all routing modes, so they are excluded from this stack-level comparison; total parameter counts (Table 5) do include them.

G Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for SMART's router, both the data-driven Mixture-of-Recursions core and the biology-informed prior of Sec. 3.

Routing as conditional computation. The router implements *conditional computation*: rather than sending every token through a fixed stack, a learned policy routes each token to a token-specific amount of compute. This is the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the “experts” here are recursion *depths* of one shared block rather than separate sub-networks, which is what couples adaptive computation (Graves 2016) to weight sharing.

Token-choice math. Let $\mathbf{h}_m \in \mathbb{R}^d$ be a marker token and $\mathbf{W}_r \in \mathbb{R}^{K\times d}$ the router. The token is scored over the K candidate depths, softmaxed, and assigned the arg-max depth:

$$\mathbf{z}_m = \tfrac{1}{\tau}\mathbf{W}_r\mathbf{h}_m, \quad \mathbf{p}_m = \alpha\,\mathrm{softmax}(\mathbf{z}_m), \quad i_m = \arg\max_i p_{m,i}, \tag{2}$$

so token m rides the shared block through steps $0, \ldots, i_m$ and exits. The realised depth is $d_m = i_m{+}1$.

Expert-choice math. Here a per-step scalar router scores the currently active tokens and a capacity c_t keeps the top-$\lceil c_t M \rceil$; survivors are the only candidates at the next step. A token's depth d_m is the deepest step it survived. Expert-choice is load-balanced by construction (each step keeps a fixed fraction), so it needs no balancing loss; token-choice does not balance itself and adds the Switch term $\mathcal{L}_{\mathrm{bal}} = K\sum_i P_i f_i$ (mean routing probability P_i times dispatch fraction f_i at depth i) (Shazeer et al. 2017). Both add the z-loss $\mathcal{L}_z = \mathbb{E}[(\mathrm{logsumexp}\,\mathbf{z})^2]$ to keep logits bounded.

The differentiable handle. The discrete choice ($\arg\max$ / top-k) has zero gradient almost everywhere, so a bare router could not learn. SMART, like MoR, keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m\, f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$ (or p_{m,i_m}). Because g_m is smooth in $\mathbf{W}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{W}_r$ is unbroken: the hard choice does the routing, the soft gate carries the gradient (a straight-through-style estimator). The biological prior of Eq. (1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Three decades of single-cell biology rest on two facts the prior encodes. (i) A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be unevenly allocated. (ii) Gene regulatory and protein-interaction networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence, and perturbing a hub propagates across the network. We operationalise "hub" as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, so a gene is central if it co-expresses with other central genes. We z-score $\mathbf{v}$ to form π. The prior thus tells the router, before any training, which genes are network hubs worth deeper computation.

Empirical-Bayes reading. Equation (1) is a log-linear prior on the routing decision: $\tilde{r}_m^{(t)} = (\text{data evidence}) + \beta_t\,\pi_m$ is the unnormalised log-score of keeping token m, with $\beta_t\pi_m$ a Gaussian-like prior mean. The anneal $\beta_t = \beta_0(1 - \text{progress})$ is shrinkage whose strength decays as data evidence accumulates, the standard empirical-Bayes behaviour: prior-dominated when the likelihood is uninformative (early training, random hidden states), data-dominated later. This is most defensible exactly in the low-signal regime (stage, node), where the likelihood alone barely separates the depths.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no phenotype label enters it. The prior therefore injects network topology, not the answer, which is why a marker-recovery or gene-discovery claim under this prior is not circular, unlike a prior built from curated cohort-specific marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma\,\mathbf{L}\tilde{\mathbf{r}}^{(t)}$, i.e. graph-Laplacian (label-propagation) regularisation: a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters, so the parameter-efficiency claim is untouched; we expose it as an option and leave its evaluation to future work.

H Token Reduction Validation

This appendix gives the full token-reduction validation summarised in Sec. 4. The protocol measures, on the headline cohort model, how effectively the marker router reduces the input token space while preserving the informative features, and is organised into four parts plus a ranking-consistency analysis. The structured record is regenerated by `python -m recursive_marker_transformer.token_reduction` and written to `token_reduction.json`; the complete per-gene ranking is written to `token_reduction_ranking.csv`.

(1) Token reduction summary. The router reduces the 4,000 gene tokens of a sample to 256 marker tokens, a 93.6% reduction (Table 21). Because self-attention is quadratic in the token count, this lowers the attention cost by a factor $(N/M)^2$ =244.1×, which is the upstream source of the parameter and FLOP savings reported in the main paper.

(2) Token selection information. The selected (retained) tokens are the 256 genes chosen by the M marker query slots at evaluation (one arg-max gene per slot); their indices and gene identifiers, and the count of discarded tokens, are stored in the record. The genes appear at the top of the ranking in Table 22.

(3) Feature importance ranking. Every gene is scored by its marker-query affinity (the pre-hardening query·key score, max over the M slots), giving a continuous ranking over all 4,000 genes. Table 22 lists both the descending view (most important first) and the ascending view (least important first); each entry is a rank, gene identifier, and importance score. The two views are exact reverses of one ranking.

(4) Selection metadata. Table 21 records the configuration of the reduction: the selection method (cross-attention marker router with recursive refinement), the importance score, the top-k value (M query slots), the selection rule (soft over all genes during training with a temperature anneal, hard arg-max at evaluation, no score threshold), and the relevant model hyperparameters.

Ranking consistency. Two quality checks accompany the reduction. First, the retained tokens are disproportionately important: their mean importance is 34.13 against 11.96 for the discarded tokens, and they hold 16.3 of the total importance mass while being a 93.6%-smaller set, so selection is far from random. Second, a gene's selection importance and the compute it is subsequently allocated are nearly uncorrelated in rank (Spearman ρ=-0.062 between marker-query affinity and mean recursion depth): *which* genes are kept and *how deeply* a kept gene is then iterated are largely independent, complementary signals rather than two views of one quantity, consistent with the main-paper finding that adaptive routing buys compute rather than accuracy. We report this rank correlation transparently rather than as a positive consistency claim.

Quantity	Value
Original gene tokens (N)	4,000
Retained marker tokens (M)	256
Removed tokens	3,744 (93.6%)
Attention-cost factor $(N/M)^2$	244.1×
Importance mass retained	16.3
Mean importance, kept vs. dropped	34.13 vs. 11.96
ρ(importance, recursion depth)	-0.062
Selection method	cross-attention marker router
Top-k / selection rule	M=256 slots, hard arg-max at eval

Table 21: Token Reduction Validation on the cohort task: reduction statistics ($N \rightarrow M$, the $O(N^2) \rightarrow O(M^2)$ attention factor), whether the kept tokens are the informative ones (importance mass and kept-vs-dropped mean importance), the importance–vs–recursion-depth rank correlation, and the selection metadata (method, top-k, selection rule).

Most important (descending)			Least important (ascending)		
Rank	Gene	Score	Rank	Gene	Score
1	ADAM12	42.58	1	ALS2CR11	6.69
2	IL1R2	42.18	2	SPINK2	6.73
3	SLC6A2	41.96	3	B3GNT3	6.90
4	SRMS	41.93	4	MAST1	7.04
5	MEG3	41.68	5	AGR3	7.05
6	CADM3	41.63	6	PRUNE2	7.22
7	CES1	41.44	7	TRY6	7.26
8	MT1F	40.63	8	MFAP4	7.34
9	RXFP1	40.15	9	DDX43	7.35
10	HEPN1	39.65	10	ZNF695	7.37
11	RBP5	39.40	11	SPC24	7.39
12	DKFZp434J0226	39.37	12	CD274	7.49
13	C8orf46	39.30	13	GPR110	7.51
14	FAM153A	39.27	14	TMPRSS11A	7.53
15	SFTPB	39.17	15	TNFSF15	7.57

Table 22: Feature importance ranking (marker-query affinity): the top most important (descending) and least important (ascending) genes, each as a rank, gene identifier, and importance score. The full per-gene ranking is in `token_reduction_ranking.csv`.